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## 1. Experiment Title:

Apply array methods and object handling; create a cart total calculator with discount logic.

## 2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

## 3. Experiment Program Code:

*Experiment 5.html*

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Shopping Cart Calculator</title>
    <link rel="stylesheet" href="Experiment 5.css">
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">
</head>
<body>
    <div class="container">
        <h2>Shopping Cart Calculator</h2>

        <div class="input-group">
            <input type="text" id="name" placeholder="Product Name">
            <input type="number" id="price" placeholder="Price">
            <input type="number" id="qty" placeholder="Quantity">
            <button onclick="addProduct()">Add Product</button>
        </div>

        <div class="table-wrapper">
            <table id="cartTable">
                <thead>
                    <tr>
                        <th>ID</th>
                        <th>Product</th>
                        <th>Price</th>
                        <th>Quantity</th>
                        <th>Total</th>
                    </tr>
                </thead>
                <tbody>
                </tbody>
            </table>
        </div>

        <div class="result-card">
            <div id="result" class="totals-summary">
                <p><b>Total Amount :</b> 0.00</p>
                <p><b>Discount :</b> 0.00</p>
```

```
                <p><b>Final Amount :</b> 0.00</p>
            </div>

            <div class="lists-container">
                <div class="summary-section">
                    <h3>Item Summary</h3>
                    <ul id="summary"></ul>
                </div>
                <div class="expensive-section">
                    <h3>Expensive Products (Price > 1000)</h3>
                    <ul id="expensive"></ul>
                </div>
            </div>
        </div>

        <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-
top: 1px solid lightgray; padding-top: 15px;">
            Made by Rohan Laharwani PRN - 24070521055
        </footer>
    </div>

    <script src="Experiment 5.js"></script>
</body>
</html>
```

### *Experiment 5.css*

```
* {
    box-sizing: border-box;
}

body {
    background-color: whitesmoke;
    color: dimgray;
    font-family: sans-serif;
    margin: 0;
    padding: 40px 20px;
    display: flex;
    justify-content: center;
}

.container {
    width: 100%;
    max-width: 700px;
    background-color: white;
    padding: 30px;
    border: 1px solid lightgray;
    border-radius: 12px;
    box-shadow: 0 4px 15px lightgray;
}

h2 {
    color: darkblue;
    text-align: center;
    margin-top: 0;
    margin-bottom: 24px;
    font-size: 24px;
}

.input-group {
    display: flex;
    gap: 12px;
    margin-bottom: 24px;
    flex-wrap: wrap;
}

input {
    flex: 1 1 140px;
```

```
    min-width: 0;
    padding: 10px 14px;
    border: 1px solid lightgray;
    border-radius: 6px;
    font-size: 14px;
    outline: none;
    transition: border-color 0.2s;
}

input:focus {
    border-color: royalblue;
}

button {
    padding: 10px 20px;
    background-color: royalblue;
    color: white;
    border: none;
    border-radius: 6px;
    font-size: 14px;
    font-weight: 600;
    cursor: pointer;
    transition: background-color 0.2s;
}

button:hover {
    background-color: blue;
}

table {
    width: 100%;
    border-collapse: collapse;
    margin-bottom: 24px;
    background-color: white;
    border-radius: 6px;
    overflow: hidden;
    border: 1px solid lightgray;
}

th, td {
    padding: 12px;
    border-bottom: 1px solid lightgray;
    text-align: center;
    font-size: 14px;
}

th {
    background-color: royalblue;
    color: white;
    font-weight: 600;
}

tr:nth-child(even) {
    background-color: whitesmoke;
}

tr:last-child td {
    border-bottom: none;
}

.result-card {
    background-color: whitesmoke;
    padding: 20px;
    border: 1px solid lightgray;
    border-radius: 8px;
    line-height: 1.8;
    font-size: 15px;
}
```

```
.totals-summary p {
    margin: 6px 0;
    font-size: 15px;
    color: #333;
}

.totals-summary p:nth-child(2) {
    color: #d9534f;
    font-weight: bold;
}

.totals-summary p:nth-child(3) {
    color: #28a745;
    font-weight: bold;
    font-size: 17px;
}

.lists-container {
    margin-top: 20px;
    border-top: 1px solid lightgray;
    padding-top: 15px;
}

h3 {
    color: darkblue;
    font-size: 16px;
    margin-top: 15px;
    margin-bottom: 8px;
}

ul {
    padding-left: 20px;
    margin-top: 4px;
    margin-bottom: 12px;
    color: dimgray;
}
```

***Experiment 5.js***

```
let cart = [];

function addProduct() {
    let name = document.getElementById("name").value.trim();
    let price = parseFloat(document.getElementById("price").value);
    let qty = parseInt(document.getElementById("qty").value);

    if (name === "" || isNaN(price) || isNaN(qty) || price <= 0 || qty <= 0) {
        alert("Please enter valid product details");
        return;
    }

    let product = {
        id: cart.length + 1,
        name: name,
        price: price,
        quantity: qty
    };

    cart.push(product);
    displayCart();

    document.getElementById("name").value = "";
    document.getElementById("price").value = "";
```

```
    document.getElementById("qty").value = "";
}

function displayCart() {
    let tableBody = document.querySelector("#cartTable tbody");
    if (!tableBody) {
        let table = document.getElementById("cartTable");
        tableBody = document.createElement("tbody");
        table.appendChild(tableBody);
    }

    tableBody.innerHTML = "";

    cart.forEach(function(item) {
        tableBody.innerHTML += `
            <tr>
                <td>${item.id}</td>
                <td>${item.name}</td>
                <td>${item.price.toFixed(2)}</td>
                <td>${item.quantity}</td>
                <td>${(item.price * item.quantity).toFixed(2)}</td>
            </tr>
        `;
    });

    let total = cart.reduce(function(sum, item) {
        return sum + (item.price * item.quantity);
    }, 0);

    let discount = 0;
    let discountPercent = 0;

    if (total >= 50000) {
        discountPercent = 20;
        discount = total * 0.20;
    } else if (total >= 20000) {
        discountPercent = 10;
        discount = total * 0.10;
    } else if (total >= 5000) {
        discountPercent = 5;
        discount = total * 0.05;
    }

    let finalAmount = total - discount;

    document.getElementById("result").innerHTML = `
        <p><b>Total Amount :</b> ₹${total.toFixed(2)}</p>
        <p><b>Discount (${discountPercent}%) :</b> -₹${discount.toFixed(2)}</p>
        <p><b>Final Amount :</b> ₹${finalAmount.toFixed(2)}</p>
    `;

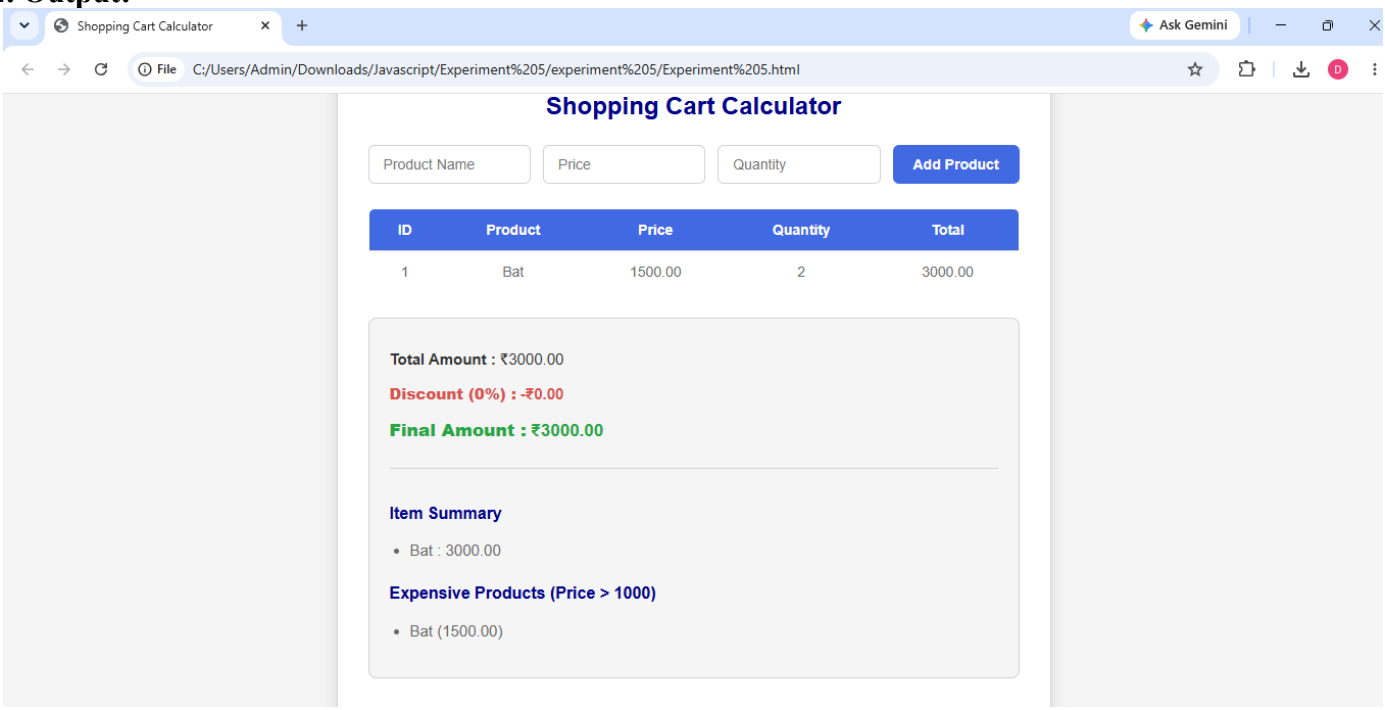
    let summary = document.getElementById("summary");
    summary.innerHTML = "";
    cart.map(function(item) {
        summary.innerHTML += `<li>${item.name} : ${(item.price * item.quantity).toFixed(2)}</li>`;
    });

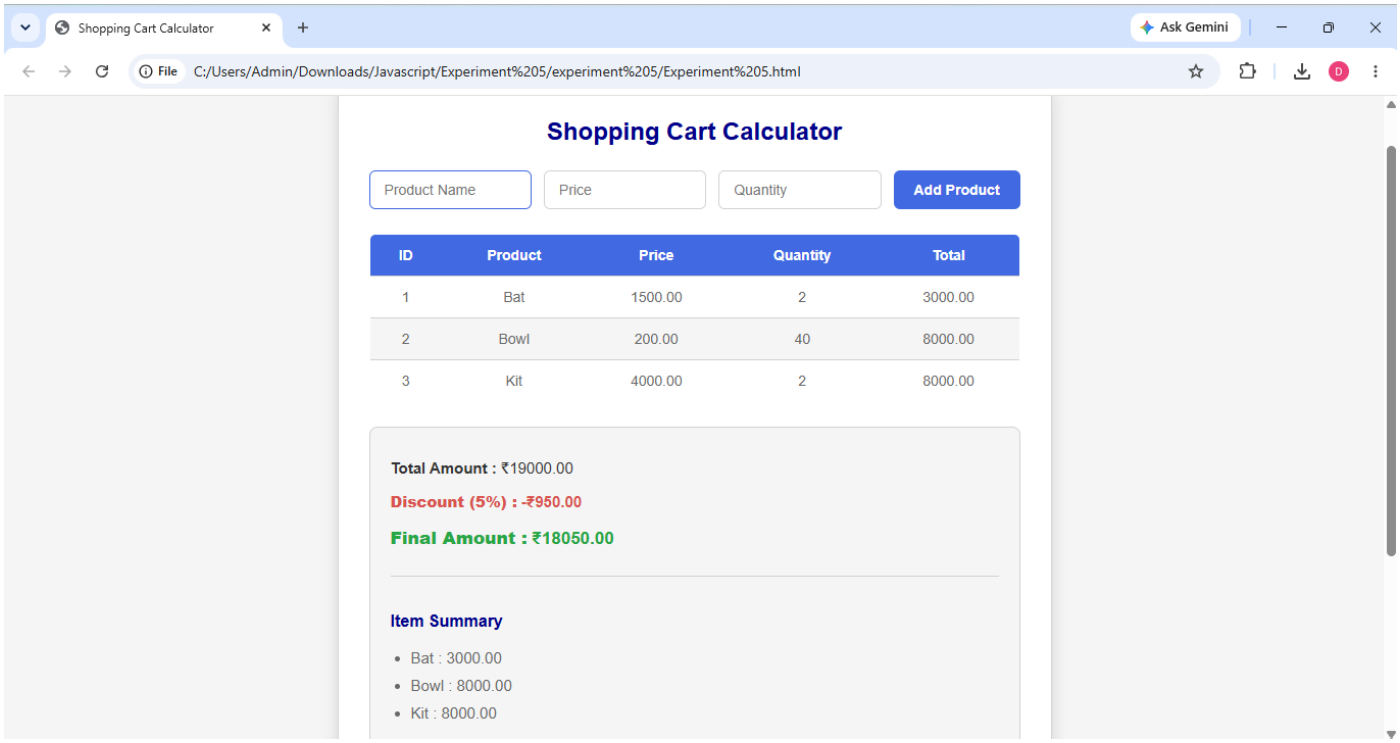
    let expensive = document.getElementById("expensive");
    expensive.innerHTML = "";

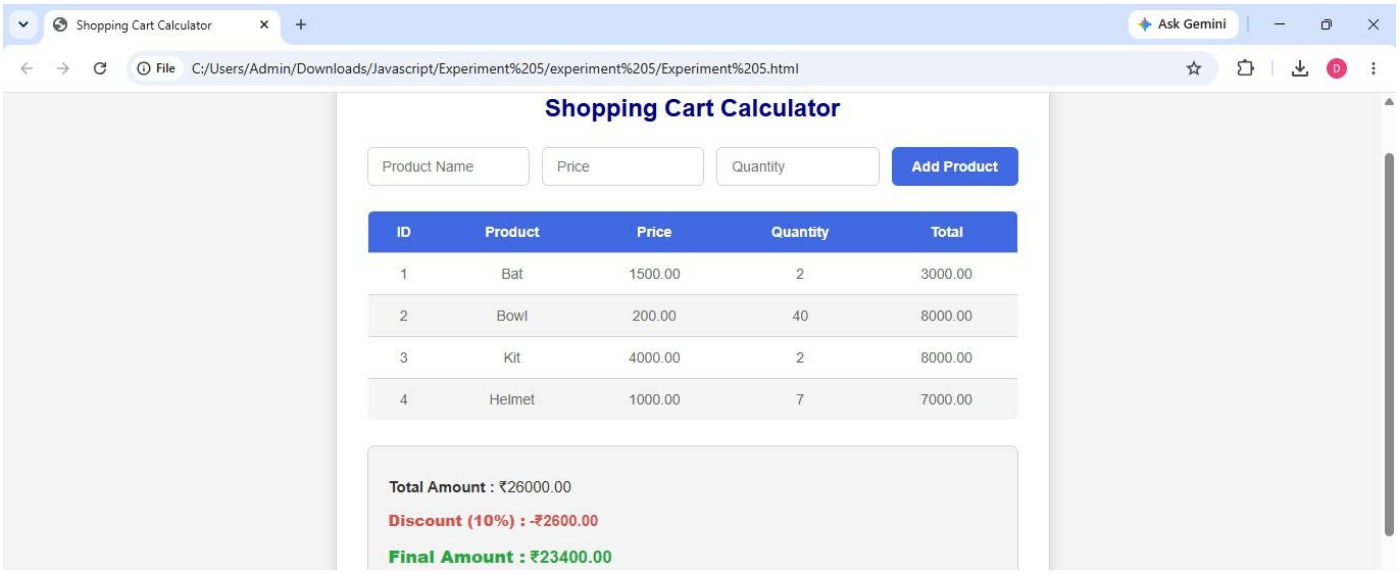
    let exp = cart.filter(function(item) {
        return item.price > 1000;
    });

    exp.forEach(function(item) {
        expensive.innerHTML += `<li>${item.name} (${item.price.toFixed(2)})</li>`;
    });
}
```

## 4. Output:


Shopping Cart Calculator
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/Experiment%205/experiment%205/Experiment%205.html
Shopping Cart Calculator
Product Name
Price
Quantity
Add Product
ID Product Price Quantity Total
1 Bat 1500.00 2 3000.00
Total Amount : ₹3000.00
Discount (0%) : -₹0.00
Final Amount : ₹3000.00
Item Summary
Bat : 3000.00
Expensive Products (Price > 1000)
Bat (1500.00)


Shopping Cart Calculator
Ask Gemini
File C:/Users/Admin/Downloads/Javascript/Experiment%205/experiment%205/Experiment%205.html
Shopping Cart Calculator
Product Name
Price
Quantity
Add Product
ID Product Price Quantity Total
1 Bat 1500.00 2 3000.00
2 Bowl 200.00 40 8000.00
3 Kit 4000.00 2 8000.00
Total Amount : ₹19000.00
Discount (5%) : -₹950.00
Final Amount : ₹18050.00
Item Summary
Bat : 3000.00
Bowl : 8000.00
Kit : 8000.00




## 5. Result/Conclusion:

A shopping cart calculator was developed using arrays of objects to store product details such as name, price and quantity. The forEach() method was used to render each cart item into an HTML table, while reduce() was used to compute the total cart amount. Conditional logic was applied on the total to calculate tiered discounts, and map() and filter() were used to generate an item-wise summary and to list products priced above a threshold respectively. This experiment reinforced the use of array methods (map, filter, reduce, forEach) and objects for managing and processing structured data in JavaScript.